The historical context of this evolution is closely tied to the geopolitical tensions of the Cold War and post-Vietnam War era, where drones became critical instruments for intelligence gathering, target observation, and battlefield reconnaissance. Nations such as the United States, Israel, and the Soviet Union invested heavily in UAV research, resulting in breakthrough platforms like the *Ryan Model 147 "Lightning Bug," IAI Scout, RQ-2 Pioneer*, and *CL-289*. These systems demonstrated that unmanned aircraft could perform complex missions reliably, safely, and repeatedly, reshaping both military strategy and aerospace engineering.

The second generation also marked the emergence of civilian and commercial drone applications, as researchers began utilizing UAVs for agriculture, disaster assessment, meteorological studies, and aerial photography. The integration of GPS, compact cameras, and miniaturized electronics made drones more accessible and multifunctional.

3.1.1 Definition

The second generation of drones refers to the era of unmanned aerial vehicle (UAV) development spanning from the 1950s to the early 2000s, characterized by the introduction of advanced radio control, onboard electronic systems, and early integration of satellite-based navigation such as GPS. This generation marked the transition from manually operated, first-generation drones to semi-autonomous aerial platforms capable of executing pre-programed missions with improved stability, range, and reliability.

Unlike their predecessors, second-generation drones incorporated digital communication links, flight control computers, and sensor-based stabilization, allowing for enhanced precision and reduced operator dependency. These UAVs were no longer confined to experimental or training roles but were deployed for real-time surveillance, reconnaissance,